

Artificial Intelligence - Presentation

Master in Artificial Intelligence and Computational Engineering 2026-27

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Universidad
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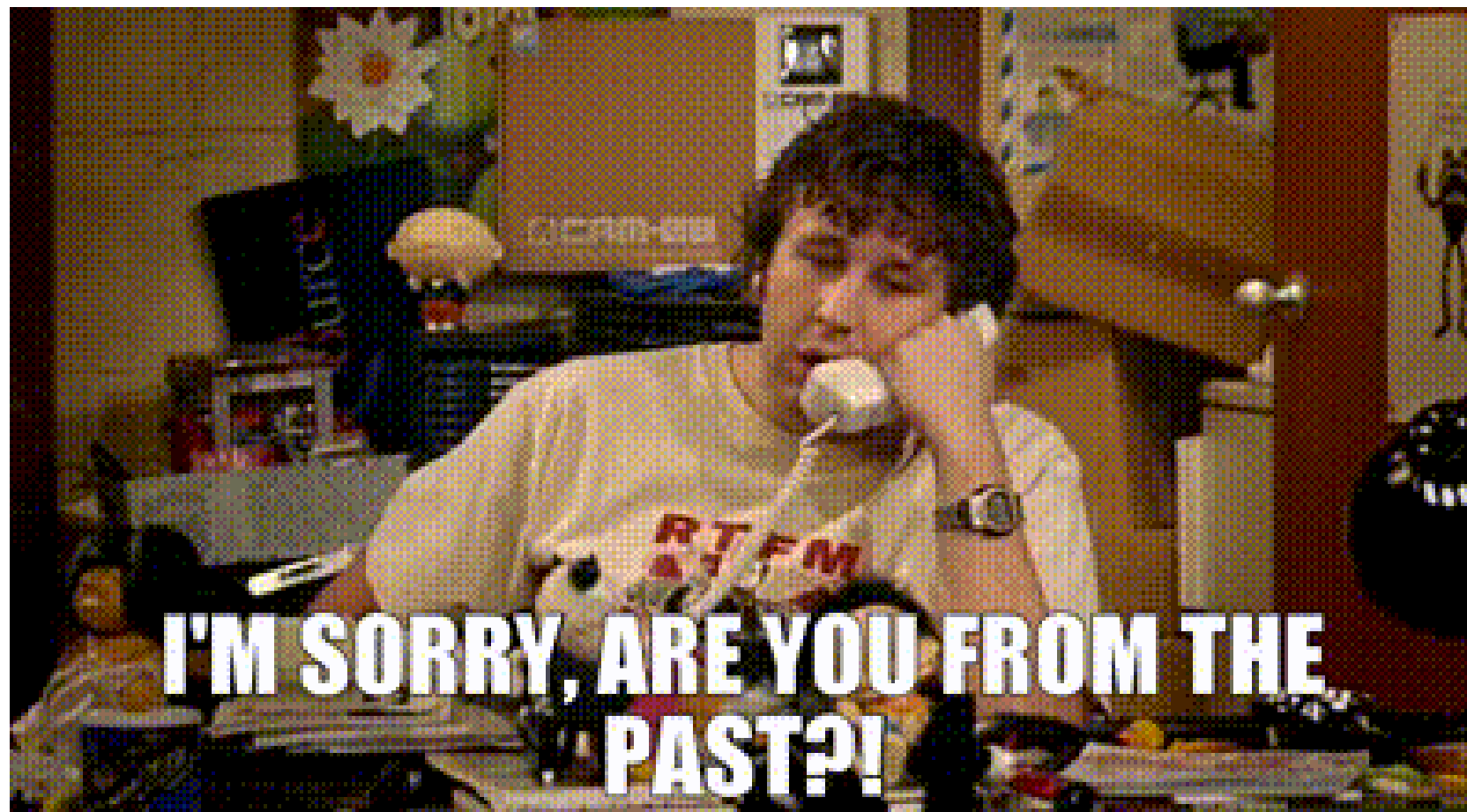
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<https://directorio.unizar.es/?#/tutoria?colectivo=PDI&codCentro=110>

Motivation



AI success

Can you introduce yourself and tell us the importance of machine learning in yourself and other applications? I'm going to use this text as an introduction in a Machine Learning course for Master students. Keep it short so I can use a screen capture.

Sure! Here's a concise version you can use as an intro slide or screen capture:

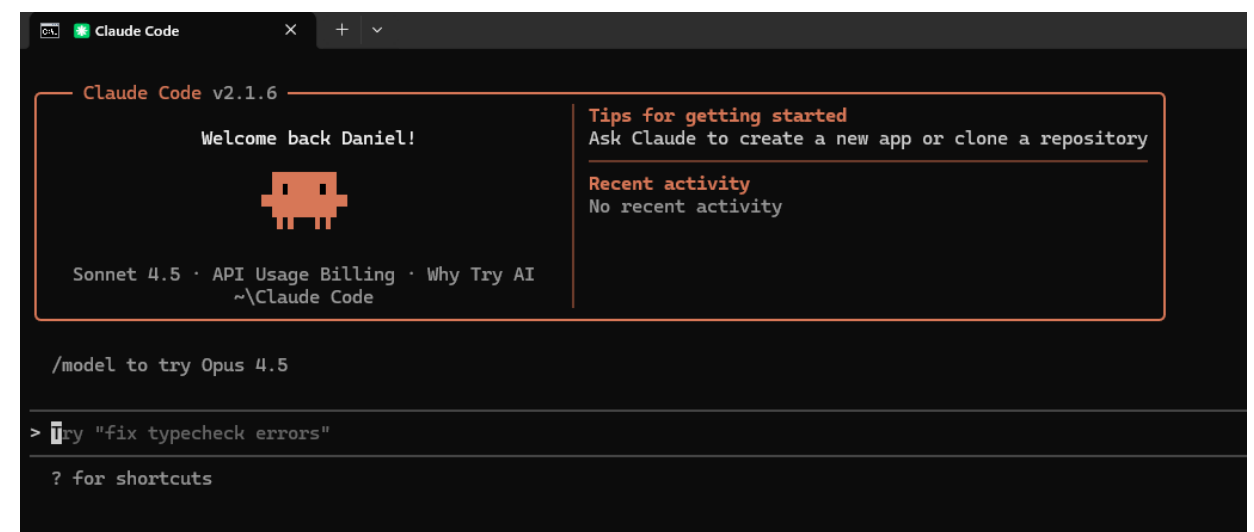
Hello, I'm **ChatGPT**, a large language model built on machine learning.

Machine learning is the foundation of my abilities—it allows me to learn patterns from vast amounts of data and generate useful, human-like responses.

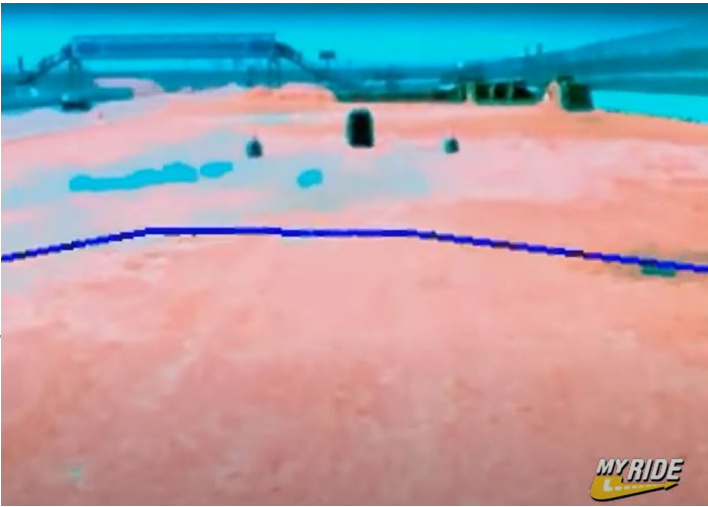
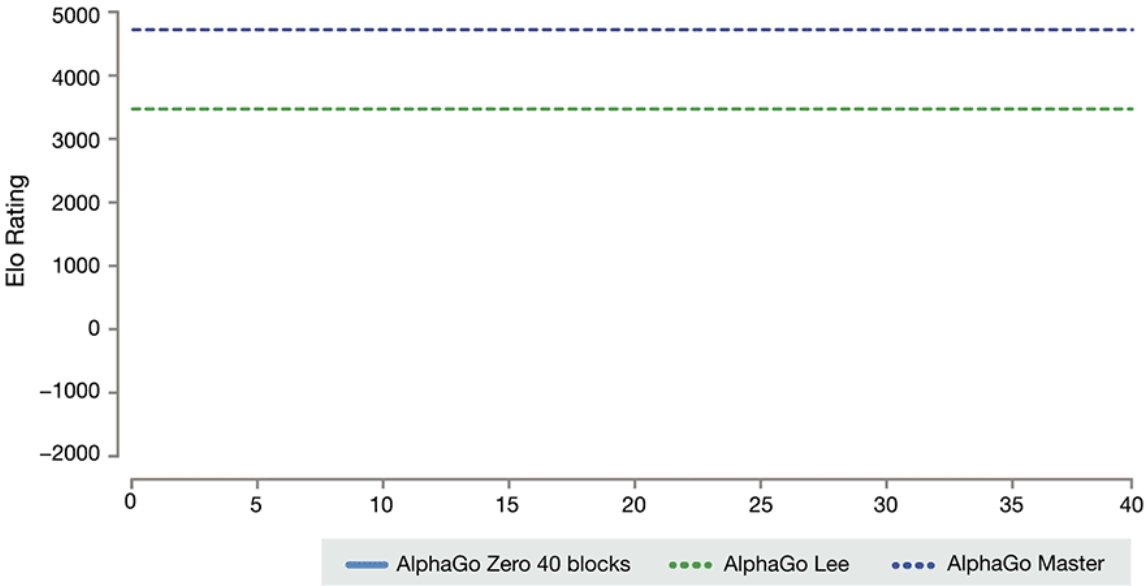
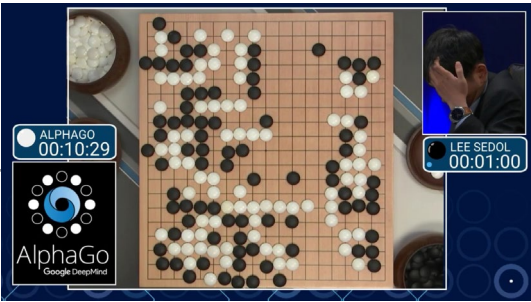
Beyond powering AI assistants, machine learning is transforming many fields:

- **Healthcare** (disease prediction, drug discovery)
- **Finance** (fraud detection, risk assessment)
- **Engineering** (predictive maintenance, optimization)
- **Everyday life** (recommendation systems, smart devices)

Machine learning matters because it enables systems to **adapt, improve, and make decisions** without being explicitly programmed for every scenario.



AI success



Stanley self driving car (2005)

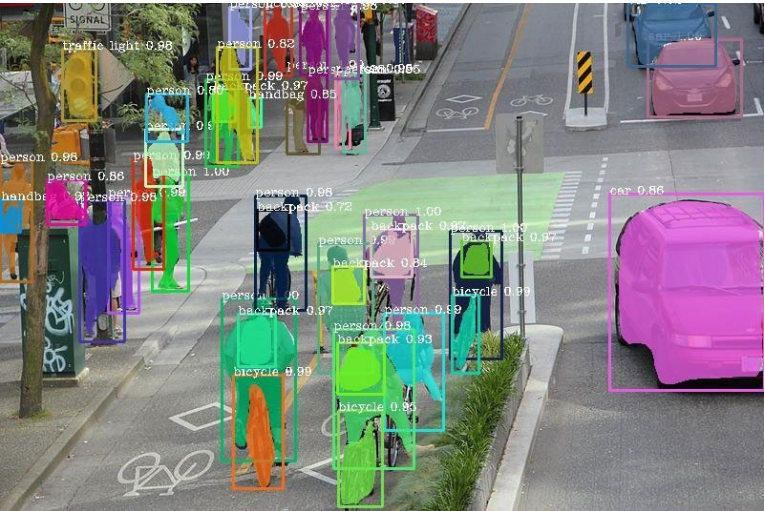
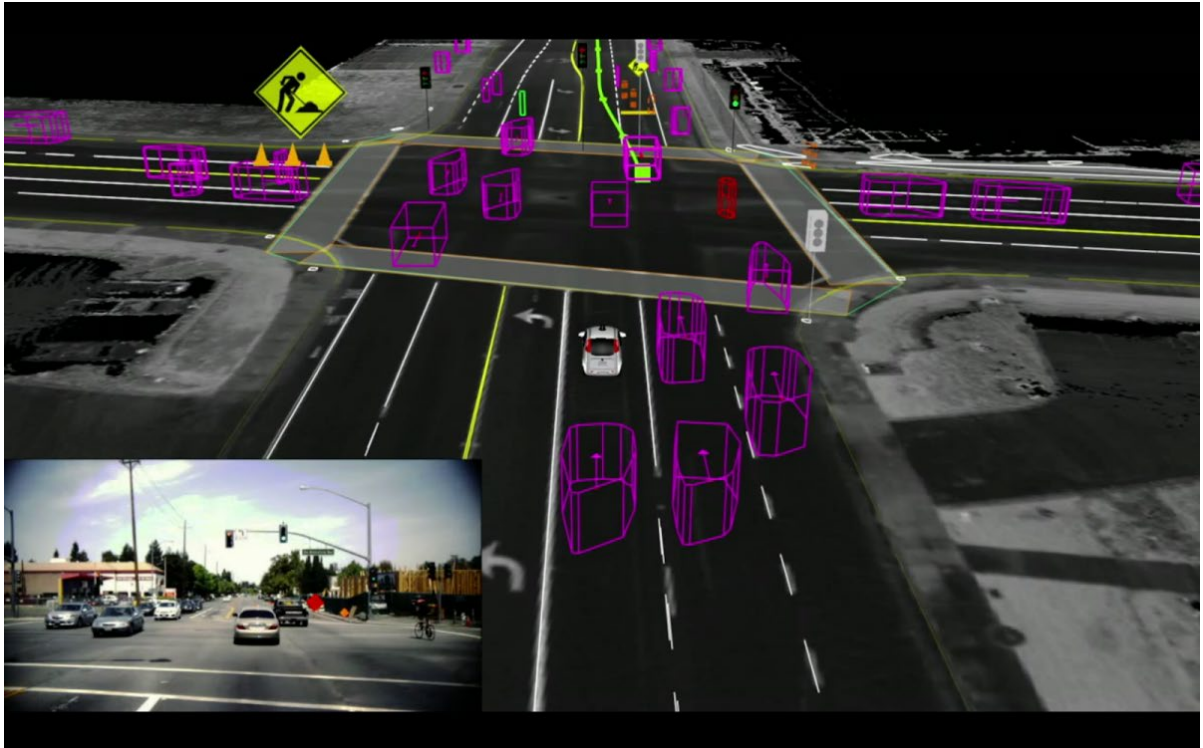


Image Segmentation

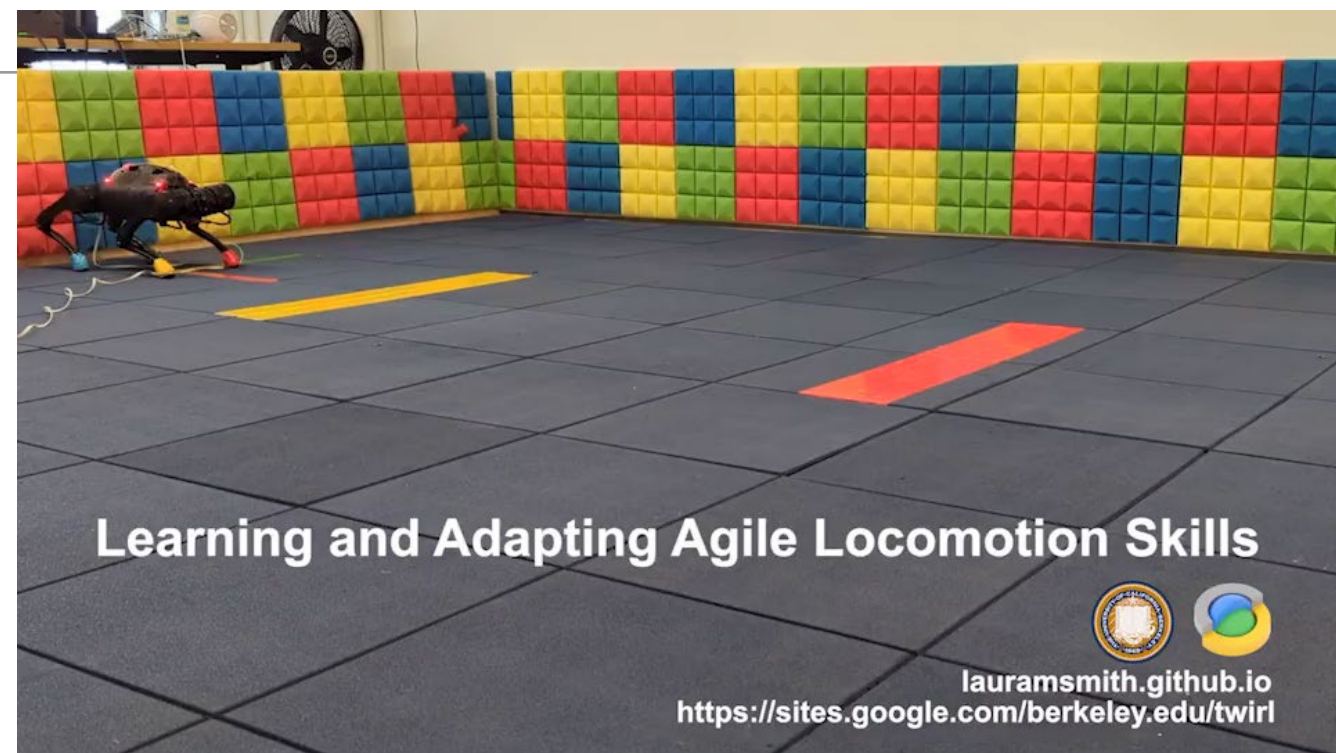
AI success



Gaussian splatting

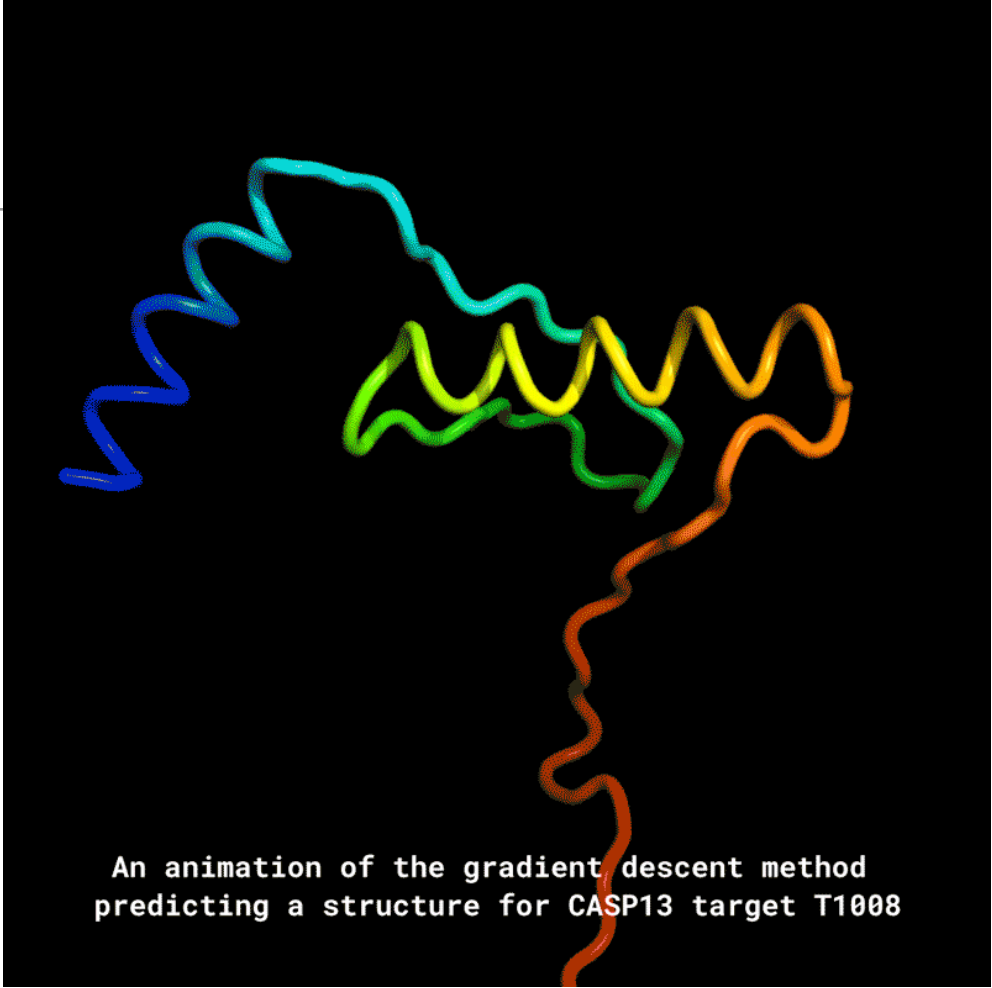
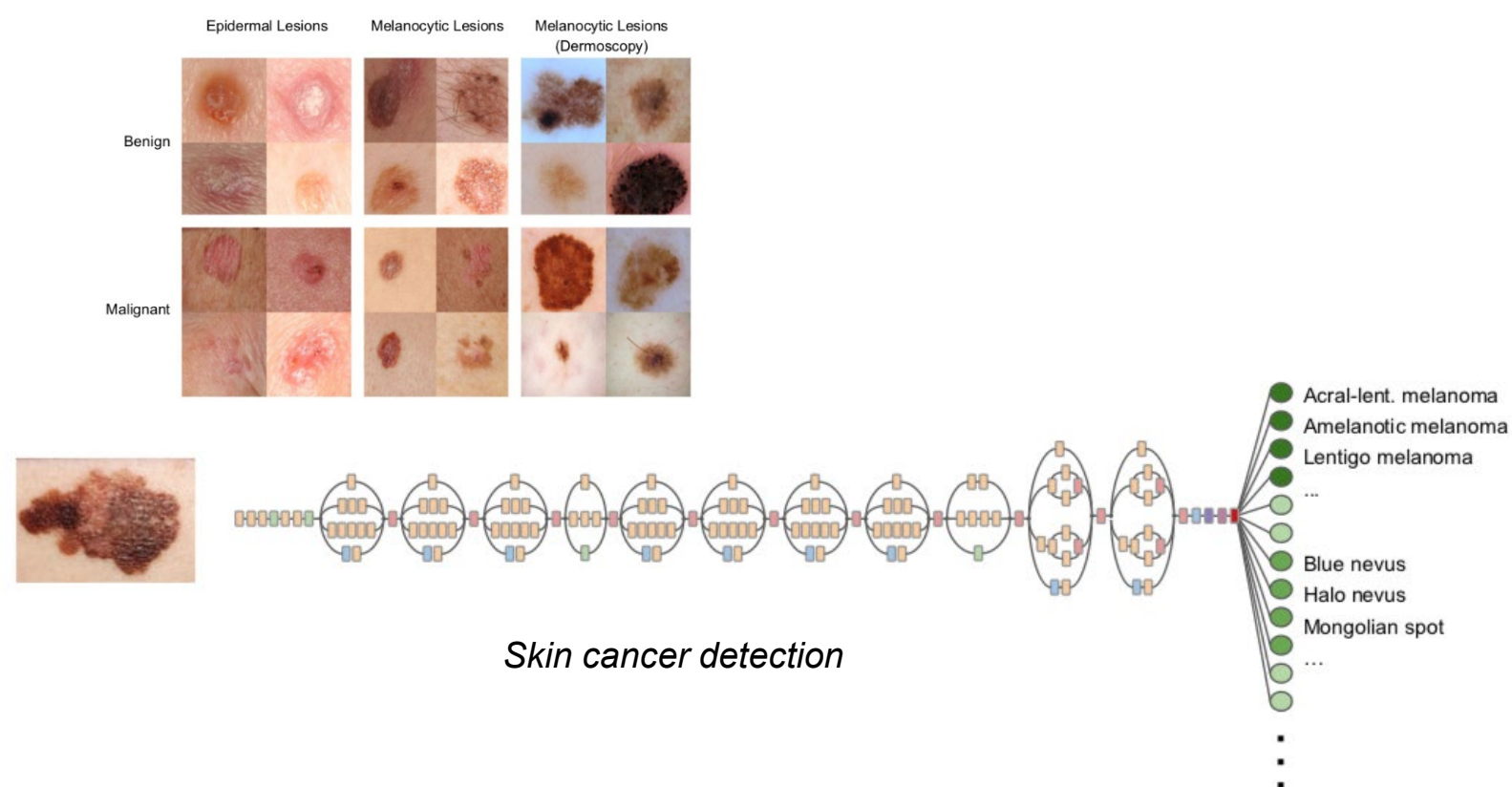


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Robots

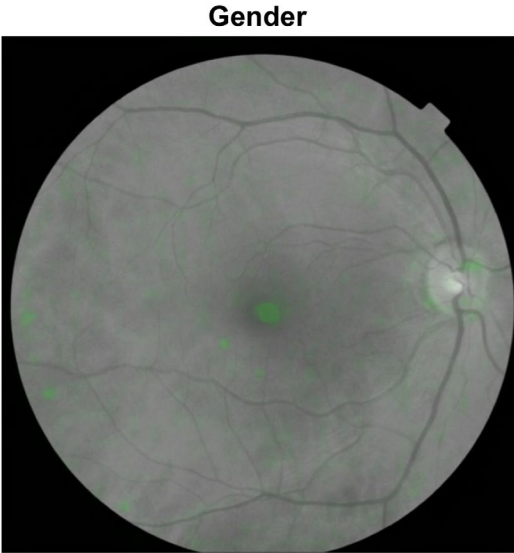
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Protein folding

Prediction of cardiovascular risk factors from retinal fundus photographs via deep learning

Table 2 Algorithm performance on predicting cardiovascular risk factors in the two validation sets				
Predicted risk factor (evaluation metric)	UK Biobank validation dataset (n = 12,026 patients)		EyePACS-2K validation dataset (n = 999 patients)	
	Algorithm	Baseline	Algorithm	Baseline
	(95% CI)		(95% CI)	
Age: MAE, years (95% CI)	3.26 (3.22,3.31)	7.06 (6.98,7.13)	3.42 (3.23,3.61)	8.48 (8.07,8.90)
Age: R ² (95% CI)	0.74 (0.73,0.75)	0.00	0.82 (0.79,0.84)	0.00
Gender: AUC (95% CI)	0.97 (0.966,0.971)	0.50	0.97 (0.96,0.98)	0.50



Actual: Female
Predicted: Female

Goals

- Study main **artificial intelligence and machine learning techniques** ,
- understand their **mathematical and algorithmic foundations** ,
- **apply them** in realistic examples.

For specific target competences, skills, and learning outcomes please go to the official university course guide (“[*guia docente*](#)”)

Tentative Program

- Introduction to Machine Learning / Shallow methods
- Bayesian and Reinforcement Learning
- Intro to Deep Learning / Convolutional Neural Networks
- Ethics and explainability
- Attention and transformers
- Other Deep Learning paradigms and strategies

Labs (3h sessions during class) form pairs

- Lab 1 - “Shallow” learning (15/9)
- Lab 2 – Bayesian+RLlearning (28/9)
- Lab 3 – DL and CNNs (6/10)
- Lab 4 – Ethics & Explainability (16/10)
- Lab 5 – Attention & Transformers (20/10)

*Tentative dates.
We'll confirm in
advance*

Submit one lab report per pair

Labs will be in **Python 3.*** (and packages):

- If you don't have any Python experience :
 - Python tutorial: <https://docs.python.org/3/tutorial/>
 - Python+Numpy tutorial: <http://cs231n.github.io/python-numpy-tutorial/>
 - A more detailed [tutorial](#) for installing it with Conda in your computer and learning it.
- For your laptop: Use Python 3, and **virtual environments** (conda, virtualenv...).
- Each lab report will include the packages needed (if any).
- In the cloud: We will also use [colab](#) (python in the browser) for labs and assignments.

*please, start
setting up your
laptop ASAP*

Evaluation (continuous)

- S1: 40% Written test .
- S2: 50% Results and reports for Lab 1, 2, 3, 4, 5.
Lab attendance is mandatory.
- S3: 10% Presentations and debates in class.

To pass the course:

```
if          S1 >=5 (out of 10) & S2 >= 5 (out of 10)
then final grade computed as:  $0.4*S1 + 0.5*S2 + 0.1*S3$ 
else min (S1,S2)
```

Methodology

- **Live lectures** . Interactive parts and in-class exercises/assignments/discussions.
- **Labs** → submit short results-form a few days after lab
- **Presentations** → presentation of labs or class discussions
- **Final exam**

Final schedule: check out the official Master/EINA calendar : -)

Evaluation (global)

If you fail to attend and submit the lab tasks during the established deadlines for the continuous evaluation, you can opt for a global single evaluation at the official evaluation date:

- S1: 40% Written test .
- S2: 60% Exam and evaluation of results and reports from Lab 1, 2, 3, 4, 5

To pass the course:

if $S1 \geq 5$ (out of 10) & $S2 \geq 5$ (out of 10)
then final grade computed as: $0.4 * S1 + 0.6 * S2$
else $\min(S1, S2)$

